
CODEBASICS RESUME PROJECT CHALLENGE #2

AtliQ Mart

Supply-chain service-level analysis:
how on-time and in-full delivery
performance fails against negotiated
targets across 35 retail customers —
and where to fix it first.

Power BI Dashboard Report · Rishab Arya ·
September 2026

ATLIQ MART · SUPPLY CHAIN SERVICE LEVELS

Executive Summary

29.0% OTIF vs 65.9% target	0 / 35 customers meet their OTIF target	−36.9 pts OTIF gap — the widest	1.7 days average lateness, max 3 days
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AtliQ Mart delivers **under half the reliability it promises**: only 29.0% of orders arrive on time and in full against a 65.9% negotiated target, and not a single one of the 35 retail customers meets its OTIF target. The failure is systemic — every city performs within 27.8–30.1% — but it breaks down into **two independent failure modes**: three high-volume chains collapse on on-time delivery, while a separate group ships on time but fills only ~40% of lines. This report documents the dashboard built to diagnose the problem and the evidence-based actions to fix it.

KPI	Grain	Actual	Target	Gap
On-Time % (OT)	Order	59.0%	86.1%	−27.1 pts
In-Full % (IF)	Order	52.8%	76.5%	−23.7 pts
On-Time In-Full % (OTIF)	Order	29.0%	65.9%	−36.9 pts
Line Fill Rate (LIFR)	Line	66.0%	n/a	n/a
Volume Fill Rate (VOFR)	Line	96.6%	n/a	n/a

Service-level KPIs, grain, actual vs target, and the gap in percentage points. LIFR and VOFR have no targets in the source data.

The Business Problem

AtliQ Mart, an FMCG manufacturer in Gujarat, supplies retail chains across **Surat, Ahmedabad, and Vadodara** and plans to expand into new metros and Tier-1 cities within two years. Several key customers have declined to renew their annual supply contracts, citing late and incomplete deliveries. Leadership asked the supply-chain analytics team for a **daily service-level tracking dashboard** — on-time, in-full, and OTIF performance for every customer against negotiated targets — to serve as the single source of truth for the renewal conversation, before the expansion replicates the problem in new cities.

What the stakeholders asked for

- On Time %, In Full %, and OTIF % against targets — split by city, then by customer.
- A customer × metric matrix with conditional formatting showing the gap vs target.
- OTIF vs target over months, drillable to weeks and days.
- A metric switcher to change which KPI the trend chart shows.
- Product insights: Line Fill Rate and Volume Fill Rate with sparklines.

Key Findings

Every figure was recomputed from the raw CSVs (57,096 order lines, 31,729 orders, Mar–Aug 2022) and verified against the dashboard's own chart exports.

1 Under half the promised reliability

OTIF is 29.0% vs a 65.9% target — and 0 of 35 customers meet their OTIF target.

2 Two independent failure modes

Mode A: the big-3 chains (Coolblue 13.7%, Acclaimed Stores 15.5%, Lotus Mart 16.3% OTIF; ~30% of all orders) collapse on **on-time** delivery (~29%). Mode B: Info Stores, Elite Mart, Sorefoz Mart and Vijay Stores (~23% of orders) ship **on time** (~72%) but fill only ~40% of lines.

3 Systemic, not regional

City OTIF spans only 27.8–30.1% — every city fails the same way, so the fix is operational, not local.

4 Shortages are small and partial

LIFR is 66.0% but VOFR is 96.6% — orders are almost complete rather than failed: a supply-planning problem, not a distribution collapse.

5 Flat but erratic

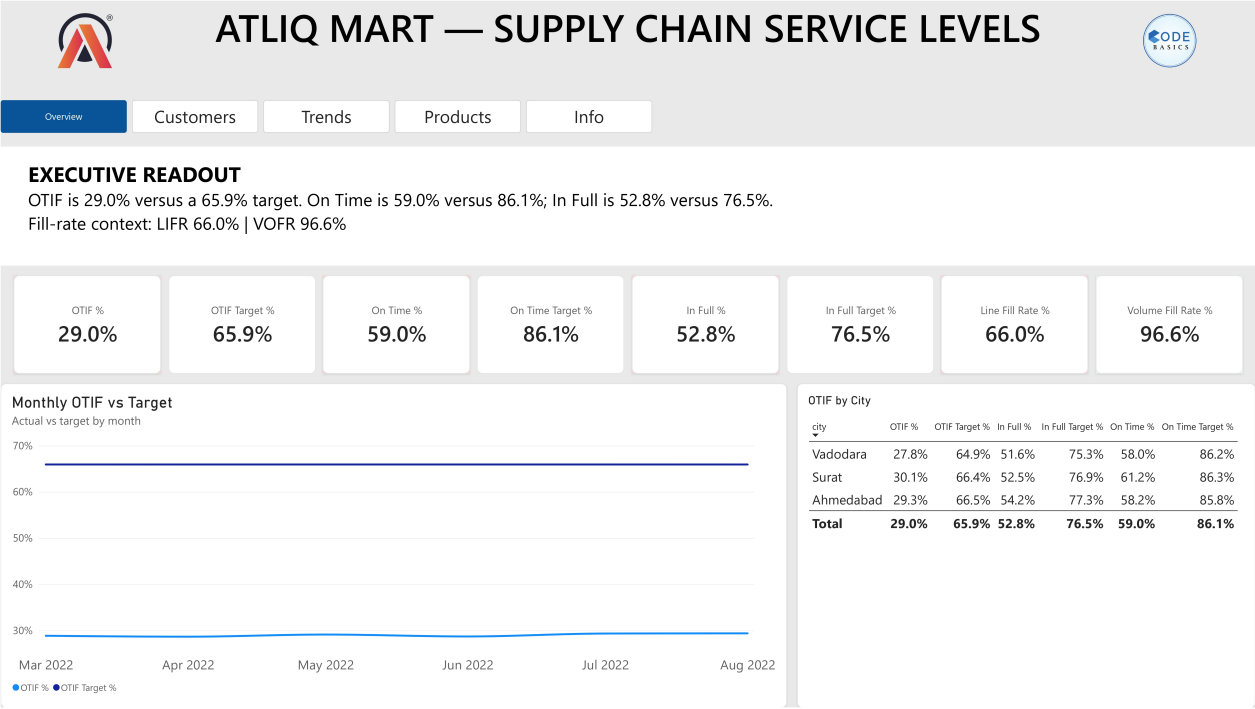
Monthly OTIF barely moves (28.7–29.4%), yet daily swings reach 22–37% — performance is chronically unstable beneath a flat average.

6 Lateness is frequent but shallow

28.9% of lines ship late, averaging 1.7 days and never more than 3 — a scheduling problem, not a transport failure.

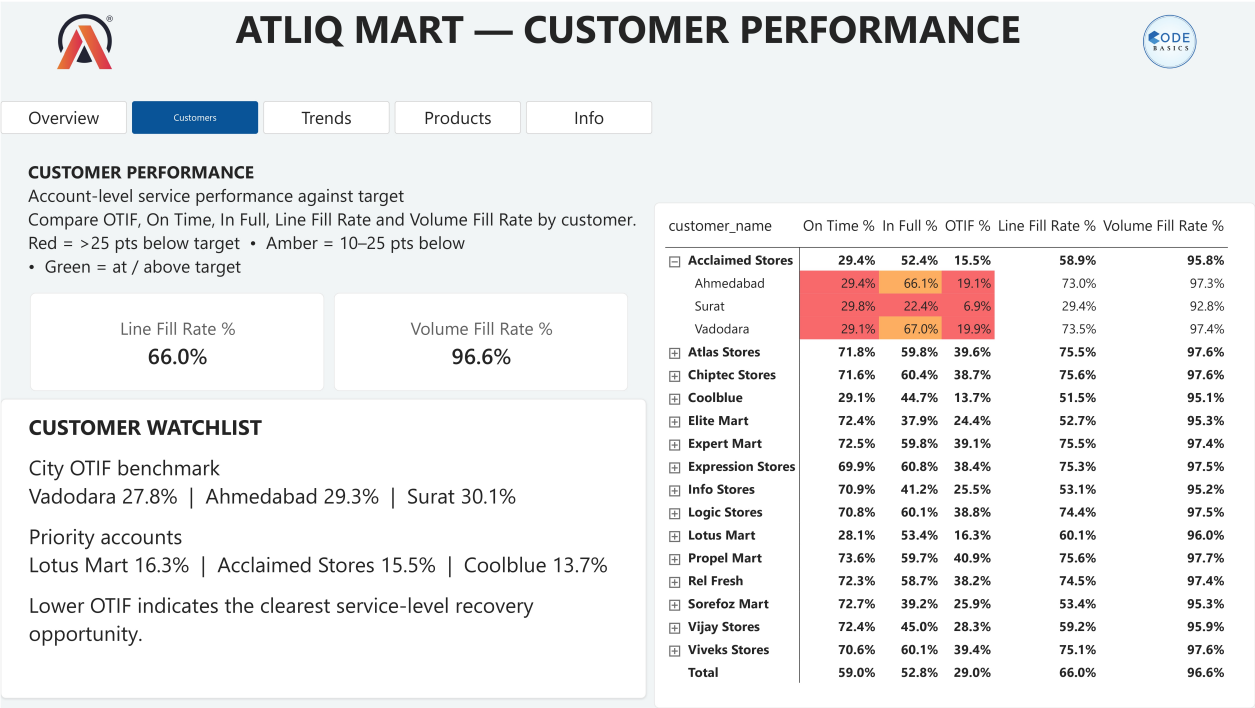
The Dashboard — Five Pages, One Story

Page 1 · Overview



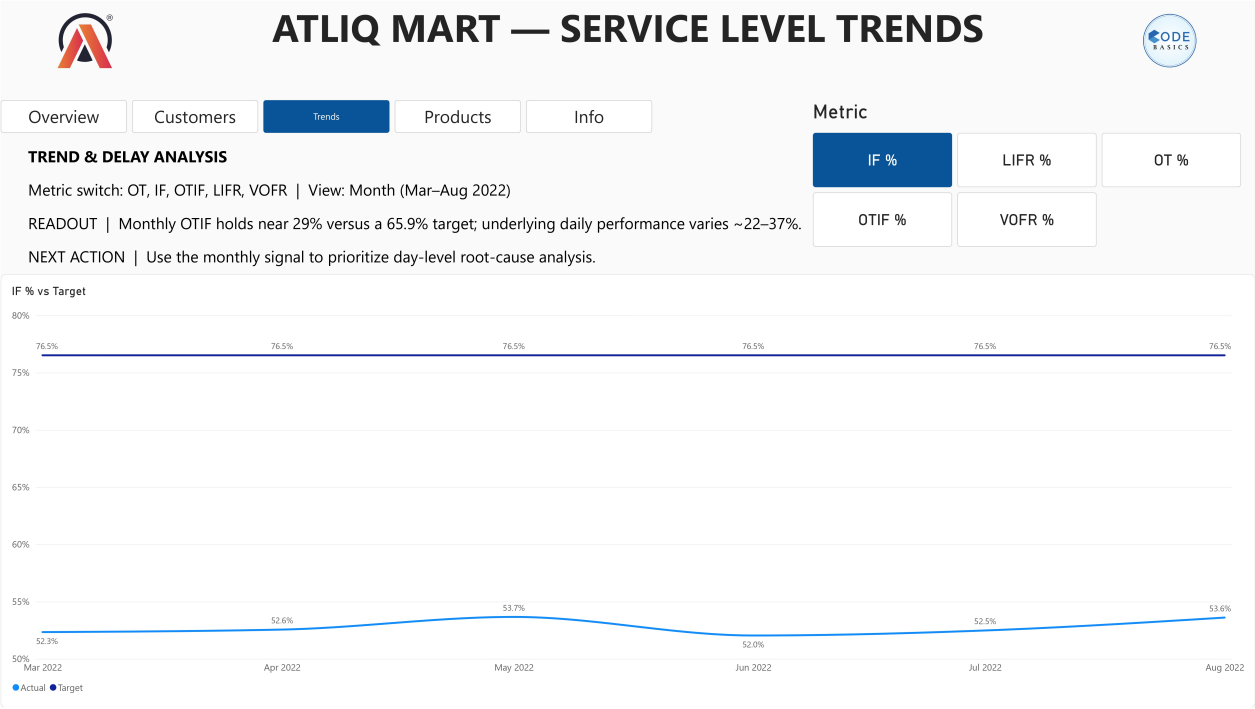
The executive landing page: all five KPIs against targets, the city split, and the monthly OTIF trend.

Page 2 · Customer Performance



Account-level matrix (OT / IF / OTIF / LIFR / VOFR) with gap-based conditional coloring, plus the city benchmark and priority-account watchlist.

Page 3 · Trends





ATLIQ MART — DEFINITIONS & DATA MODEL



OverviewCustomersTrendsProductsInfo

Key Metrics & Definitions

On Time % (OT %) — % of orders delivered on or before the agreed date. Order-level.
In Full % (IF %) — % of orders where ALL lines shipped complete. Order-level.
OTIF % — % of orders BOTH on time AND in full. The reliability metric.
Line Fill Rate % — % of order lines shipped in full quantity. Line-level.
Volume Fill Rate % — Total qty shipped ÷ total qty ordered. Line-level.

Targets: per-customer targets from dim_targets_orders (OT, IF, OTIF only — LIFR and VOFR have no targets in the source data).

Gap = Actual – Target, in percentage points. Matrix colors: green ≥ target, yellow within –10, orange within –25, red beyond –25.

Data Model (Star Schema)

Fact tables: fact_order_lines (57k order lines), fact_orders_aggregate (32k orders)
Dimensions: dim_customers (35 accounts), dim_products (18 SKUs), dim_date (Mar–Aug 2022), dim_targets_orders (per-customer targets)
Relationships: All dimensions filter facts (single direction).
Date connects via order_placement_date.
fact-to-fact joins deliberately avoided (star schema discipline).
Data cleaning: Power Query — date parsing, header fixes (BOM), category capitalization, column renames, month_sort helper column.

Key Insights

— OTIF 29.0% vs 65.9% target: under half promised reliability across all 6 months
— LIFR 66.0% but VOFR 96.6%: shortages are small and partial (supply planning issue)
— Failure is uniform across cities (OTIF 27.8–30.1%): systemic, not regional
— Monthly results are flat but daily swings reach 22–37%: chronic and erratic
— Lowest OTIF accounts: Lotus Mart (16.3%), Acclaimed Stores (15.5%), Coolblue (13.7%)
— Lowest LIFR products: AM Butter 250 (63.5%), AM Tea 250 and AM Biscuits 250 (65.2%)

Notes

— Data window: 01-Mar-2022 to 31-Aug-2022
— Product view includes native LIFR and VOFR sparklines by date
— Metric switch and drill: trend chart supports months → weeks → days

Metric definitions, the star-schema model, data-cleaning notes, and the key insights.

Recommendations

- Run **two parallel workstreams**: dispatch/scheduling for the on-time mode, supply planning for the in-full mode — they are statistically independent problems.
- Open the renewal conversation with the **big-3 chains first**: ~30% of order volume and the clearest on-time collapse (OT ~29%).
- Add a **1–2 day scheduling buffer** and review dispatch planning — lateness is frequent but shallow (max 3 days), so this is a planning fix, not a transport redesign.
- Hold **SKU-level safety stock** and review pick accuracy for the small per-line shortfalls behind the LIFR–VOFR gap (66.0% vs 96.6%).
- Pilot fixes in **Vadodara** — the weakest OTIF (27.8%) and highest-volume city.

Data Model & DAX Measures

Star schema: two fact tables — **fact_orders_aggregate** (order grain, 31,729 rows) and **fact_order_lines** (line grain, 57,096 rows) — connected to four dimensions (customers, products, date, targets) through six single-direction relationships. No fact-to-fact joins: the auto-detected order_id bridge was removed so every filter has exactly one path. Nineteen explicit DAX measures power the report; the core of them:

```
Total Orders = COUNTROWS ( fact_orders_aggregate )
```

```

Total Order Lines = COUNTROWS ( fact_order_lines )

On Time % = DIVIDE ( SUM ( fact_orders_aggregate[on_time] ), [Total Orders] )
In Full % = DIVIDE ( SUM ( fact_orders_aggregate[in_full] ), [Total Orders] )
OTIF % = DIVIDE ( SUM ( fact_orders_aggregate[otif] ), [Total Orders] )

Line Fill Rate % = DIVIDE ( SUM ( fact_order_lines[in_full] ), [Total Order Lines] )
Volume Fill Rate % = DIVIDE ( SUM ( fact_order_lines[delivery_qty] ),
SUM ( fact_order_lines[order_qty] ) )

On Time Target % = DIVIDE ( AVERAGE ( dim_targets_orders[ontime_target] ), 100 )
In Full Target % = DIVIDE ( AVERAGE ( dim_targets_orders[infull_target] ), 100 )
OTIF Target % = DIVIDE ( AVERAGE ( dim_targets_orders[otif_target] ), 100 )

OT Gap = ( [On Time %] - [On Time Target %] ) * 100
IF Gap = ( [In Full %] - [In Full Target %] ) * 100
OTIF Gap = ( [OTIF %] - [OTIF Target %] ) * 100

```

```

OT Gap Color =
VAR GapPoints = [OT Gap]
RETURN
SWITCH ( TRUE (),
GapPoints >= 0, "#63BE7B", -- green (at/above target)
GapPoints >= -10, "#FFEB84", -- yellow (within 10 pts)
GapPoints >= -25, "#FDAE61", -- orange (within 25 pts)
"#F8696B" -- red (beyond 25 pts)
)

Selected Metric Value =
SWITCH ( SELECTEDVALUE ( Metrics[Metric] ),
"OT %", [On Time %],
"IF %", [In Full %],
"OTIF %", [OTIF %],
"LIFR %", [Line Fill Rate %],
"VOFR %", [Volume Fill Rate %],
BLANK ()
)

Chart Title =
VAR MetricName = SELECTEDVALUE ( Metrics[Metric] )
VAR DrillLevel =
SWITCH ( TRUE (),

```

```

ISFILTERED ( dim_date[date] ), "Daily",
ISFILTERED ( dim_date[week_no] ), "Weekly",
ISFILTERED ( dim_date[mmm_yy] ), "Monthly",
""
)
RETURN
IF ( NOT HASONEVALUE ( Metrics[Metric] ), "Select a Metric",
IF ( DrillLevel = "", MetricName & " vs Target",
MetricName & " vs Target – " & DrillLevel ) )

```

The full 19-measure reference, including the gap-color ladders and the target router, ships with the repository in docs/MEASURES.md.

About This Project

Codebasics Resume Project Challenge #2 (AtliQ Mart). Built in Power BI Desktop: Power Query for cleaning and shaping, a six-relationship star schema, 19 DAX measures, and a five-page report. The data window is 01-Mar-2022 to 31-Aug-2022; all figures in this report were verified against the raw data.

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